

Sharpening Spatial Filters

○ Foundation

- The first-order derivative

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

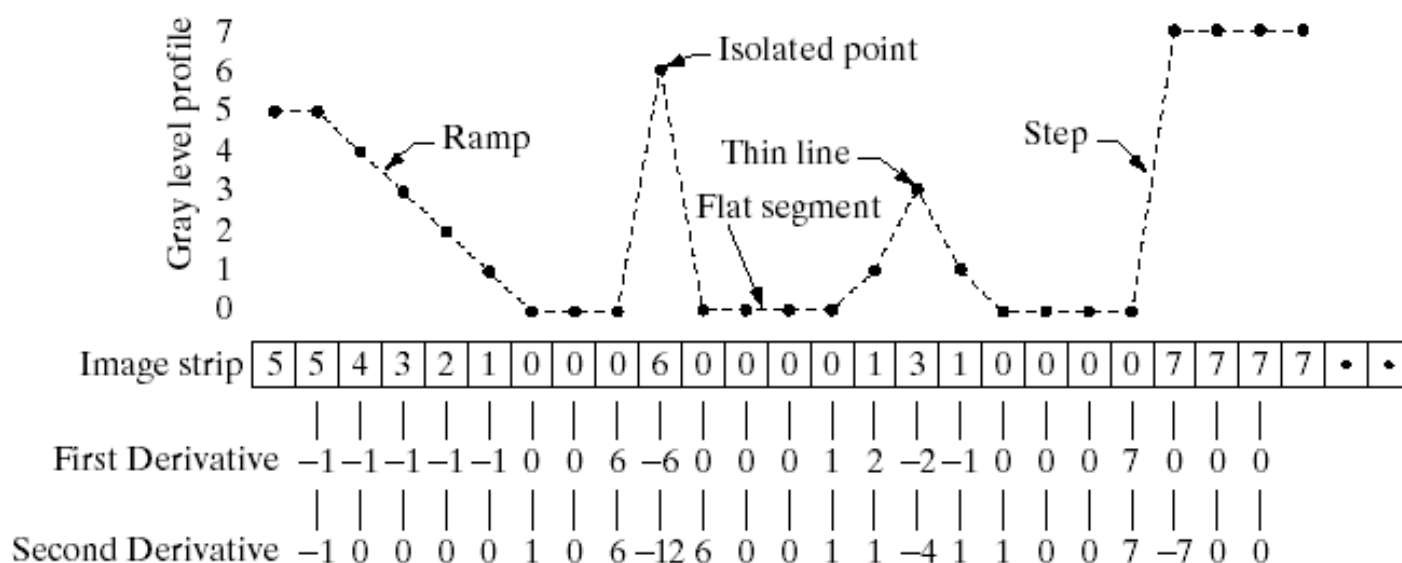
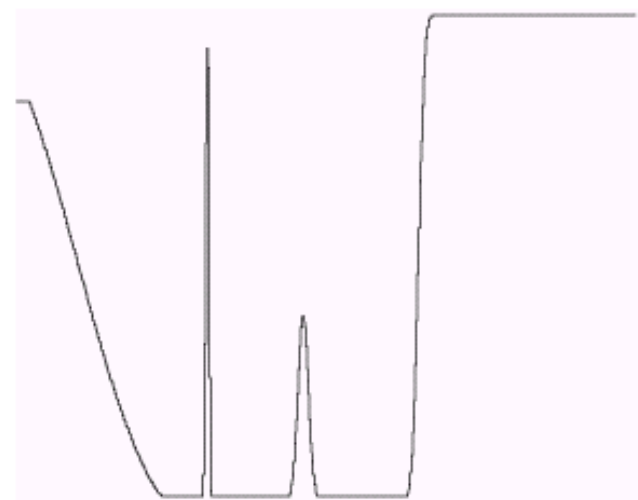
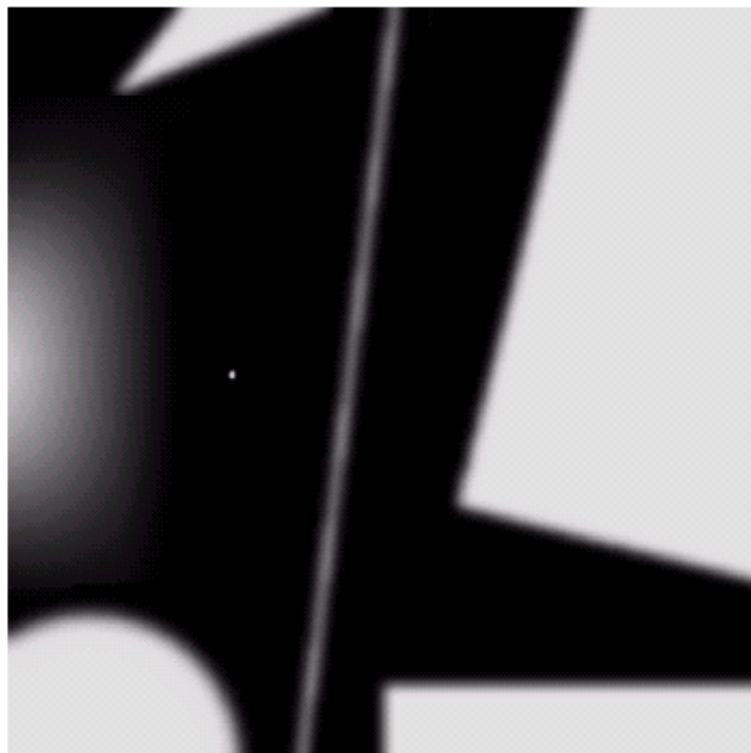
- The second-order derivative

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

a b
c

FIGURE 3.38

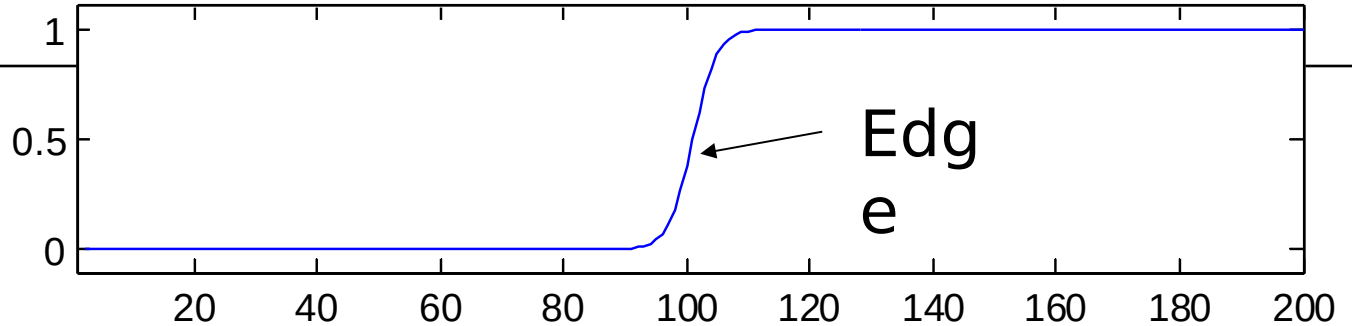
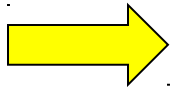
(a) A simple image. (b) 1-D horizontal gray-level profile along the center of the image and including the isolated noise point. (c) Simplified profile (the points are joined by dashed lines to simplify interpretation).



Laplacian Sharpening : How it works

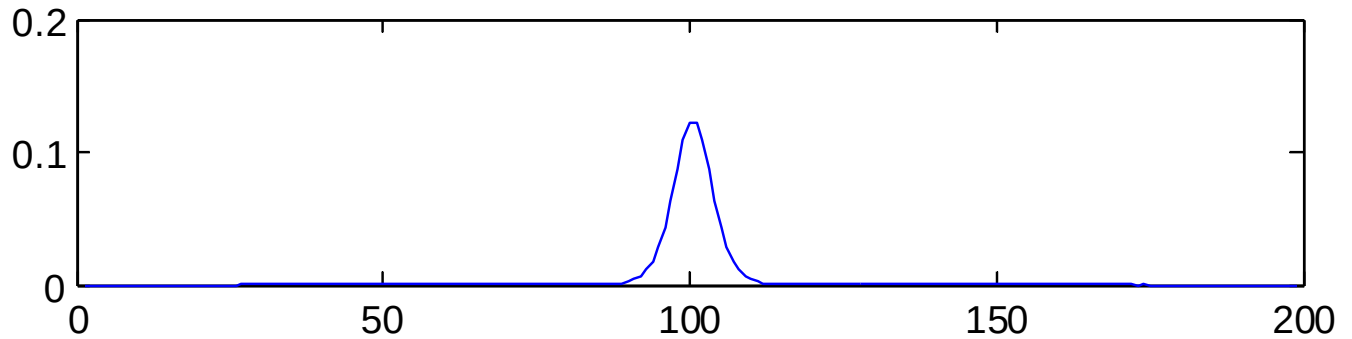
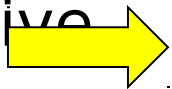
Intensity
profile

$p(x)$



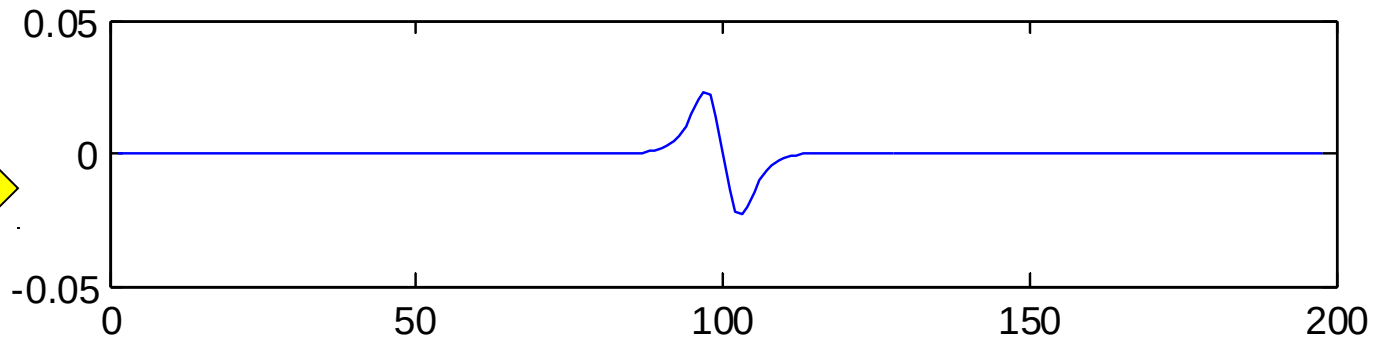
1st
derivative

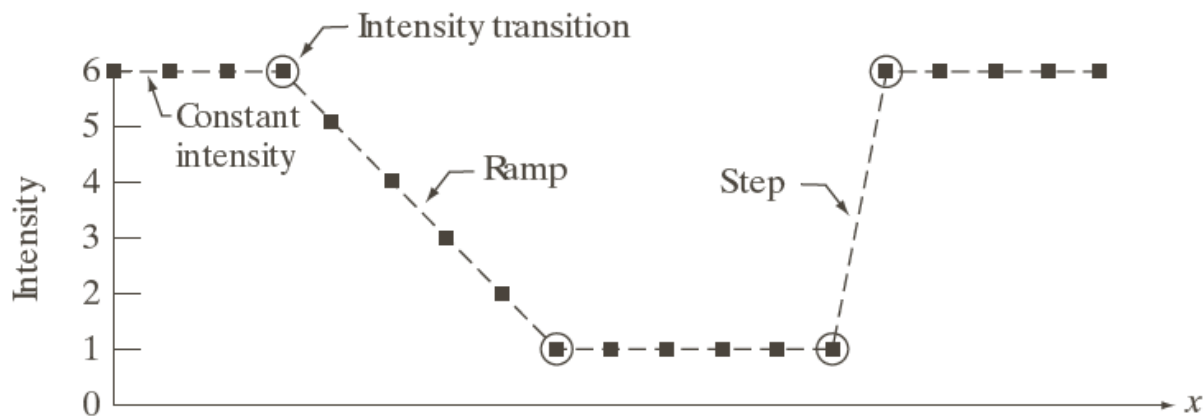
$\frac{dp}{dx}$



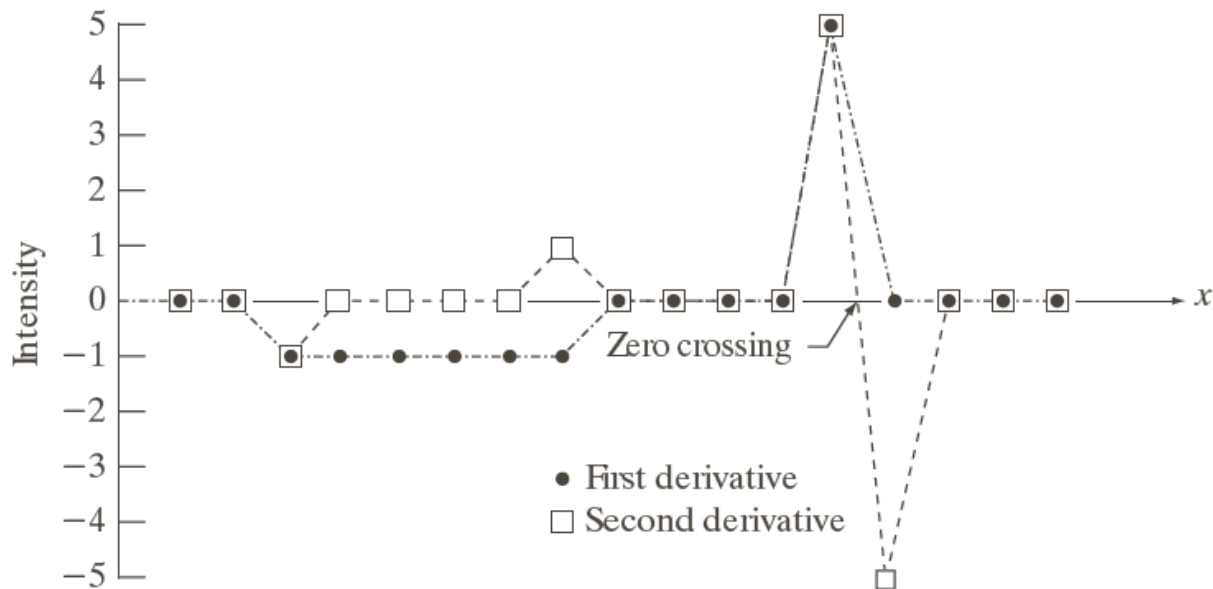
2nd
derivative

$\frac{d^2p}{dx^2}$





Scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
1st derivative	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0	5	0	0	0	0	0
2nd derivative	0	0	-1	0	0	0	0	1	0	0	0	0	0	5	-5	0	0	0	0



a
b
c

FIGURE 3.36 Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

-
- Use of second derivatives for enhancement-The Laplacian
 - Development of the method

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient} \\ & \text{of the Laplacian mask} \\ & \text{is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient} \\ & \text{of the Laplacian mask} \\ & \text{is positive} \end{cases}$$

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

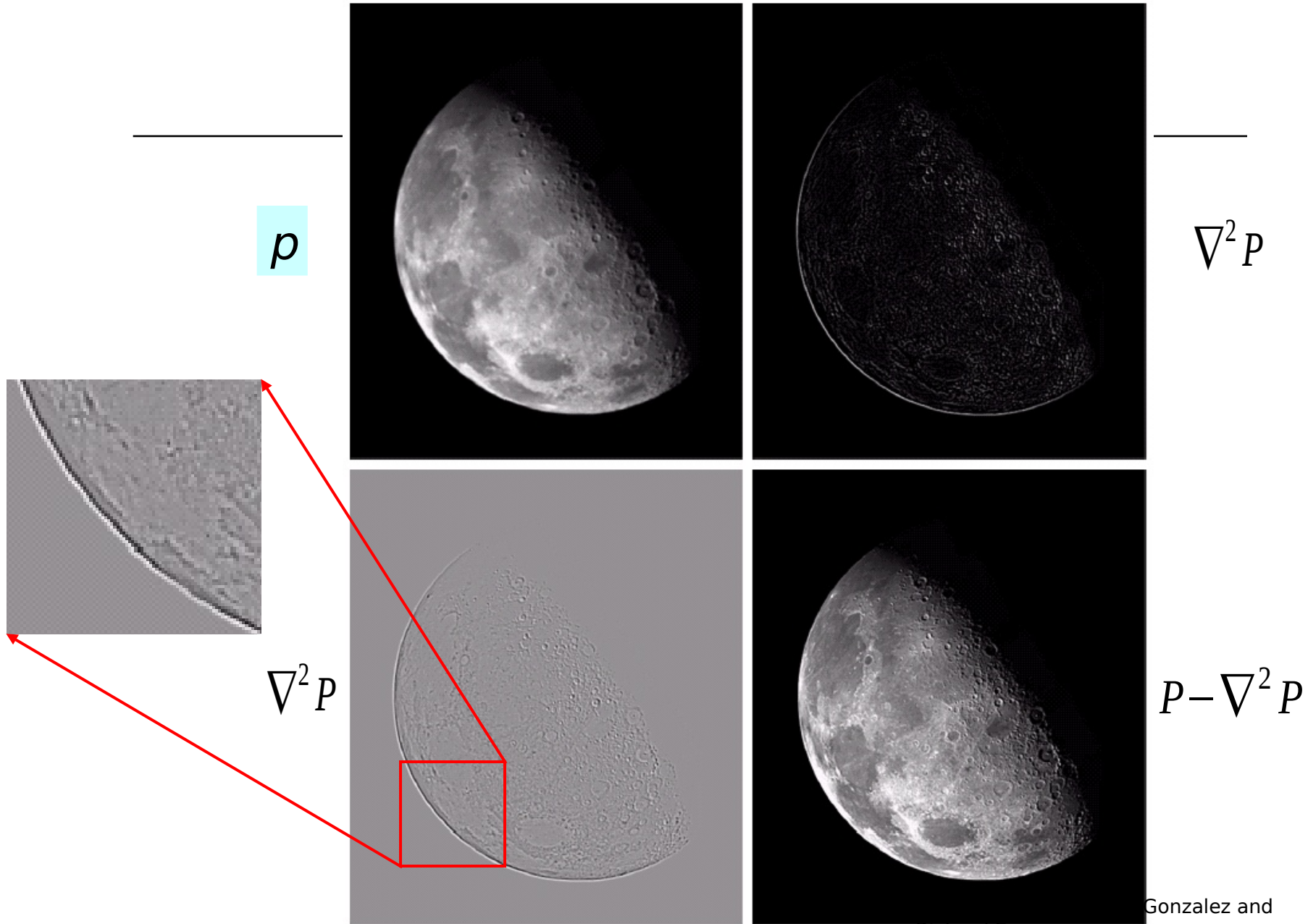
a	b
c	d

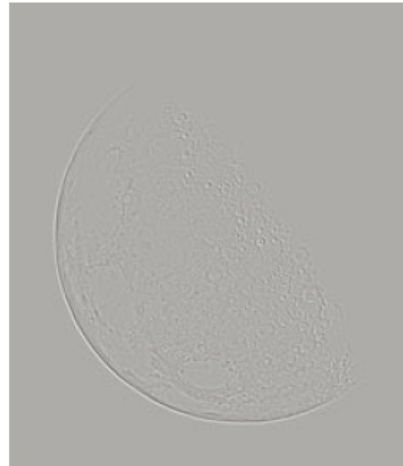
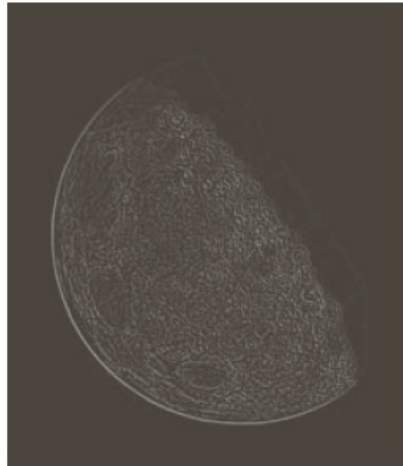
FIGURE 3.39

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4).

(b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

Laplacian Sharpening Example





a	
b	c
d	e

FIGURE 3.38

(a) Blurred image of the North Pole of the moon.

(b) Laplacian without scaling.

(c) Laplacian with scaling.

(d) Image sharpened using the mask in Fig. 3.37(a).

(e) Result of using the mask in Fig. 3.37(b).

(Original image courtesy of NASA.)

- Simplifications

$$\begin{aligned} g(x,y) &= f(x,y) - [f(x+1,y) + f(x-1,y) + f(x,y+1) + \\ &\quad f(x,y-1)] + 4f(x,y) \\ &= 5f(x,y) - [f(x+1,y) + f(x-1,y) + f(x,y+1) + \\ &\quad f(x,y-1)] \end{aligned}$$

Laplacian Sharpening (cont.)

Mask

for

$$P - \nabla^2 P$$

0	-1	0
-1	5	-1
0	-1	0

or

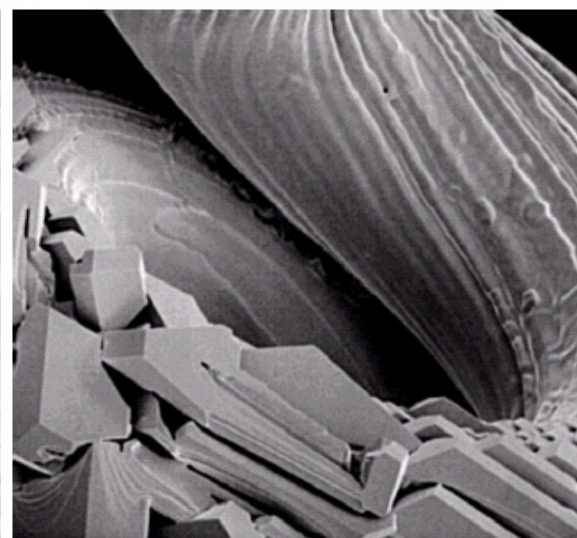
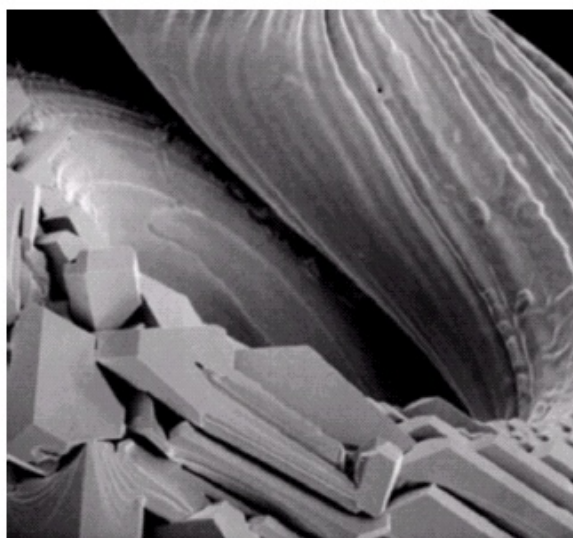
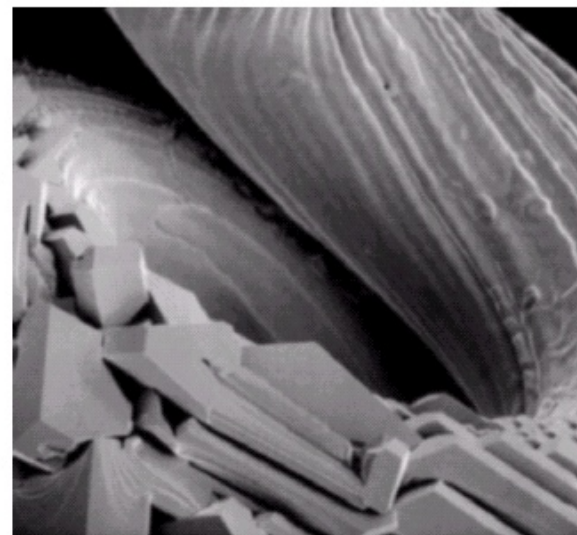
-1	-1	-1
-1	9	-1
-1	-1	-1

Mask
for $\nabla^2 P$

1	1	1
1	-8	1
1	1	1

or

0	1	0
1	-4	1
0	1	0



a b c
d e

FIGURE 3.41 (a) Composite Laplacian mask. (b) A second composite mask. (c) Scanning electron microscope image. (d) and (e) Results of filtering with the masks in (a) and (b), respectively. Note how much sharper (e) is than (d). (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Illinois at Urbana-Champaign. Images from Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, 2nd Edition, Prentice Hall, 1992.)

- Unsharp masking and highboost filtering

- Unsharp masking

- Subtract a blurred version of an image from the image itself

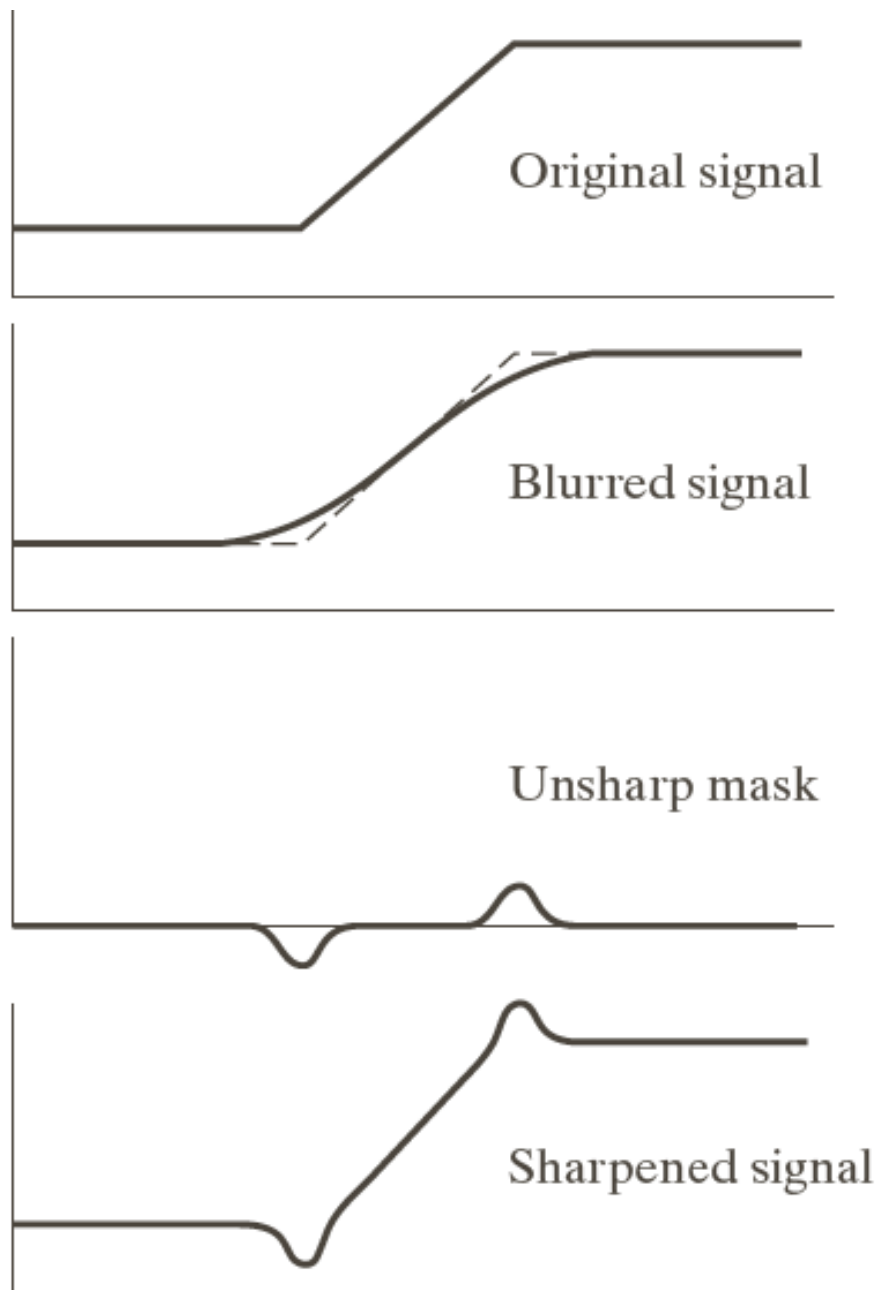
$$g_{mask}(x,y) = f(x,y) - \bar{f}(x,y)$$

- $f(x,y)$: The image, $\bar{f}(x,y)$: The blurred image

$$g(x,y) = f(x,y) + k * g_{mask}(x,y) \quad , k=1$$

-
- High-boost filtering

$$g(x,y) = f(x,y) + k * g_{mask}(x,y) \quad , k > 1$$



a
b
c
d

FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).



DIP-XE

DIP-XE

DIP-XE

DIP-XE

DIP-XE

a

b

c

d

e

FIGURE 3.40

(a) Original image.

(b) Result of blurring with a Gaussian filter.

(c) Unsharp mask. (d) Result of using unsharp masking.

(e) Result of using highboost filtering.

-
- Using first-order derivatives for (nonlinear) image sharpening—The gradient

$$|\nabla P| = \sqrt{\left(\frac{\partial P}{\partial x}\right)^2 + \left(\frac{\partial P}{\partial y}\right)^2}$$

-
- The magnitude is rotation invariant (isotropic)

$$|\nabla P| = \sqrt{\left(\frac{\partial P}{\partial x}\right)^2 + \left(\frac{\partial P}{\partial y}\right)^2}$$

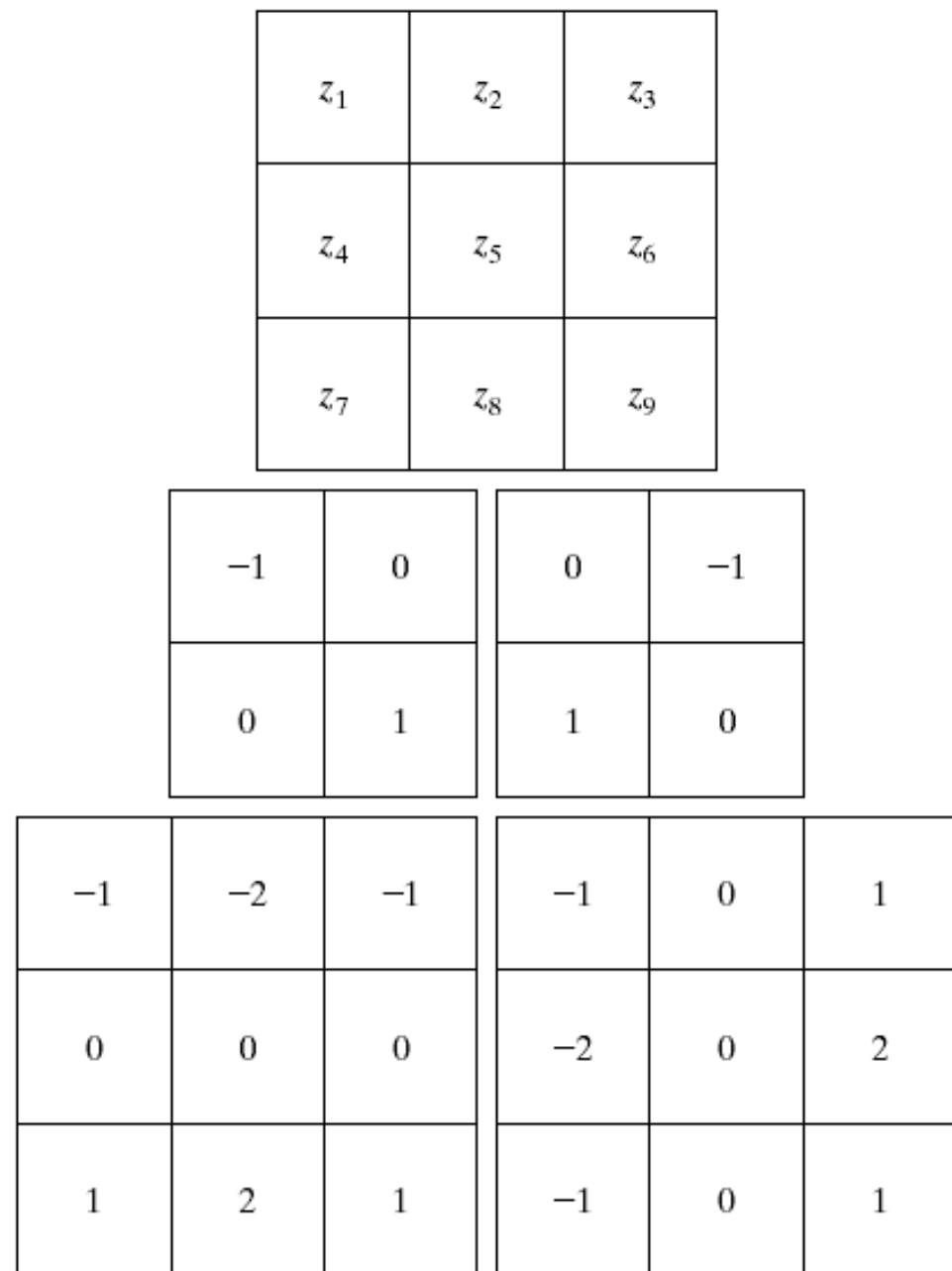
$$\nabla P \approx |G_x| + |G_y|$$

$$G_y = \frac{\partial P}{\partial y}$$

$$G_x = \frac{\partial P}{\partial x}$$

a
b c
d e

FIGURE 3.44
 A 3×3 region of an image (the z 's are gray-level values) and masks used to compute the gradient at point labeled z_5 . All masks coefficients sum to zero, as expected of a derivative operator.



-
- Computing using cross differences, Roberts cross-gradient operators

$$G_x = (z_9 - z_5) \quad \text{and} \quad G_y = (z_8 - z_6)$$

$$\nabla P = \left[(z_9 - z_5)^2 + (z_8 - z_6)^2 \right]^{1/2}$$

$$\nabla P \approx |z_9 - z_5| + |z_8 - z_6|$$

- Sobel operators

- A weight value of 2 is to achieve some smoothing by giving more importance to the center point

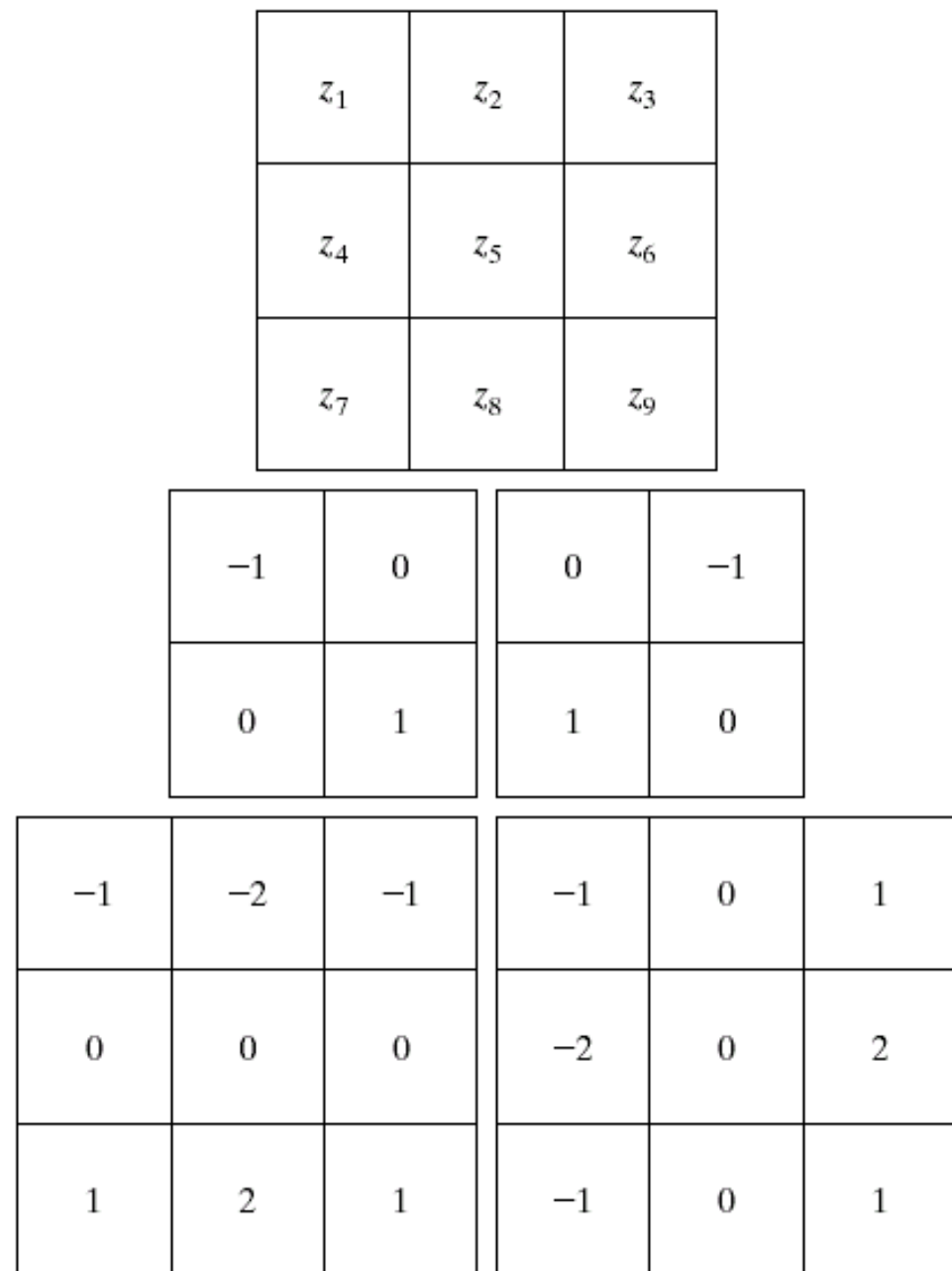
$$G_x = \left| (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3) \right|$$

$$G_y = \left| (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7) \right|$$

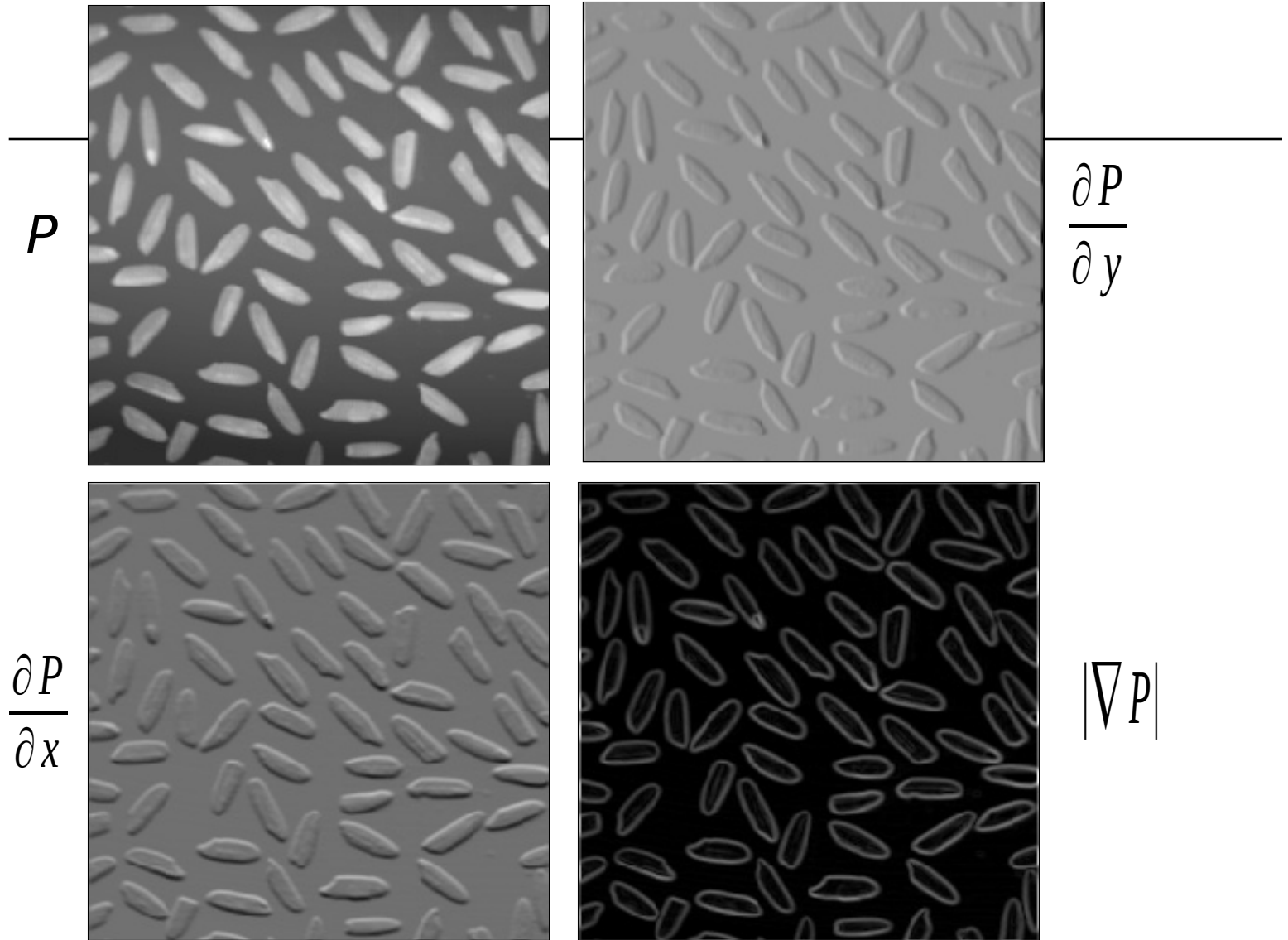
$$\nabla P \approx |G_x| + |G_y|$$

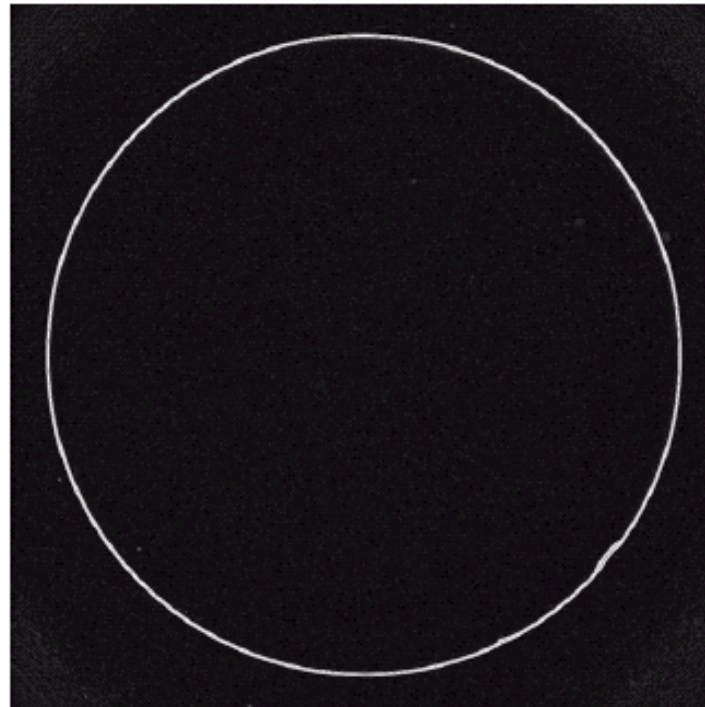
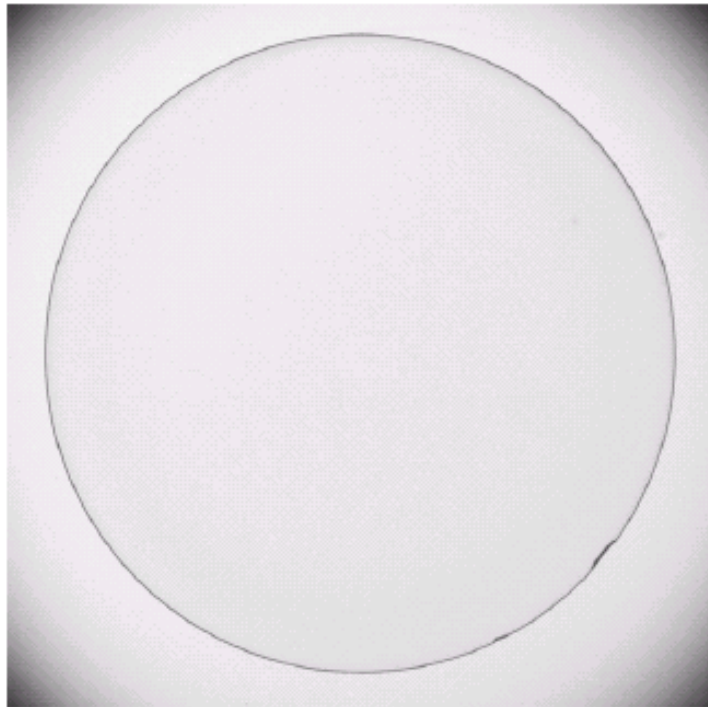
a
b c
d e

FIGURE 3.44
 A 3×3 region of an image (the z 's are gray-level values) and masks used to compute the gradient at point labeled z_5 . All masks coefficients sum to zero, as expected of a derivative operator.



First Order Partial Derivative: Image Gradient





a b

FIGURE 3.45
Optical image of
contact lens (note
defects on the
boundary at 4 and
5 o'clock).
(b) Sobel
gradient.
(Original image
courtesy of
Mr. Pete Sites,
Perceptics
Corporation.)

Image Enhancement in the Spatial Domain : *Mix things up !*

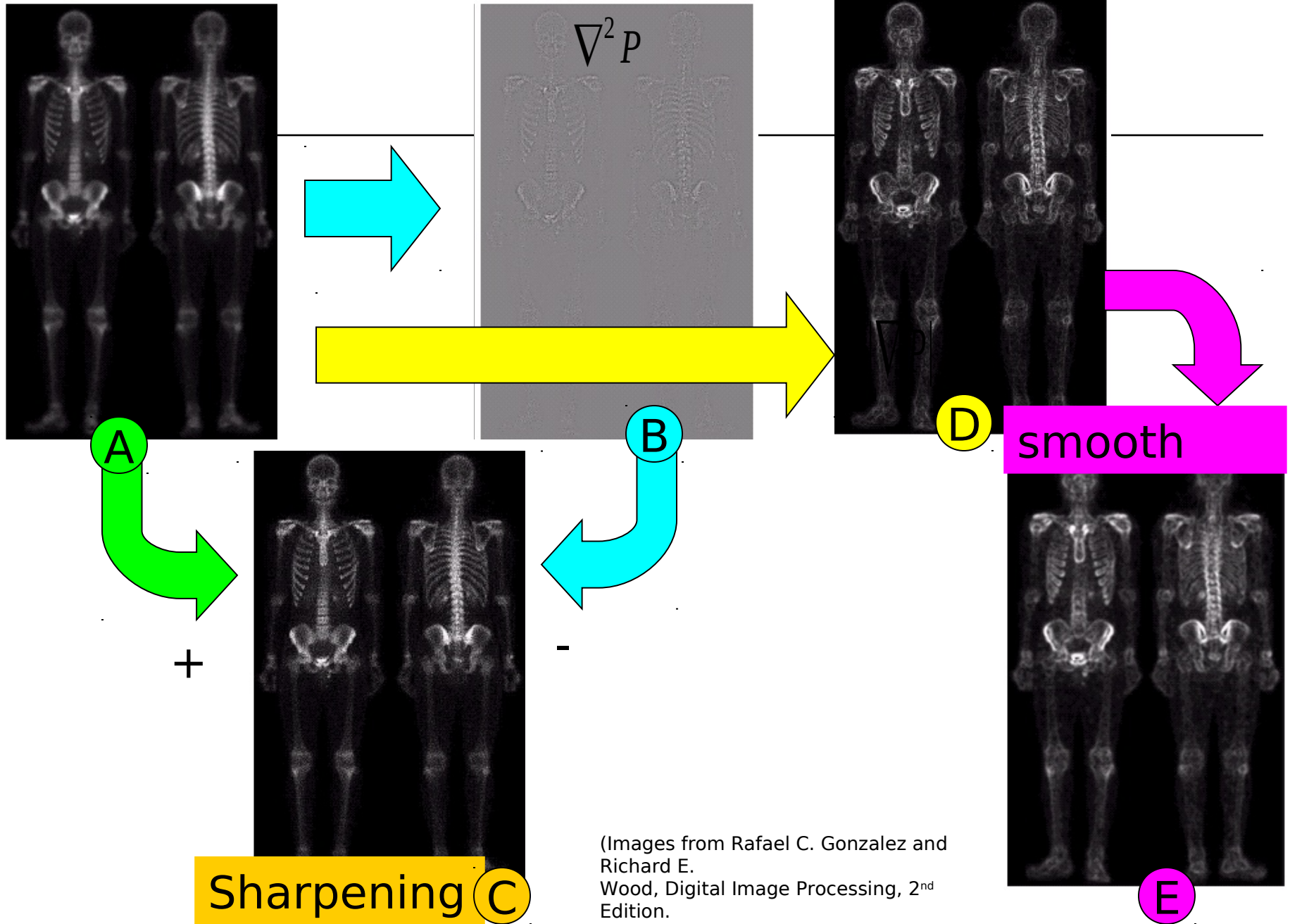
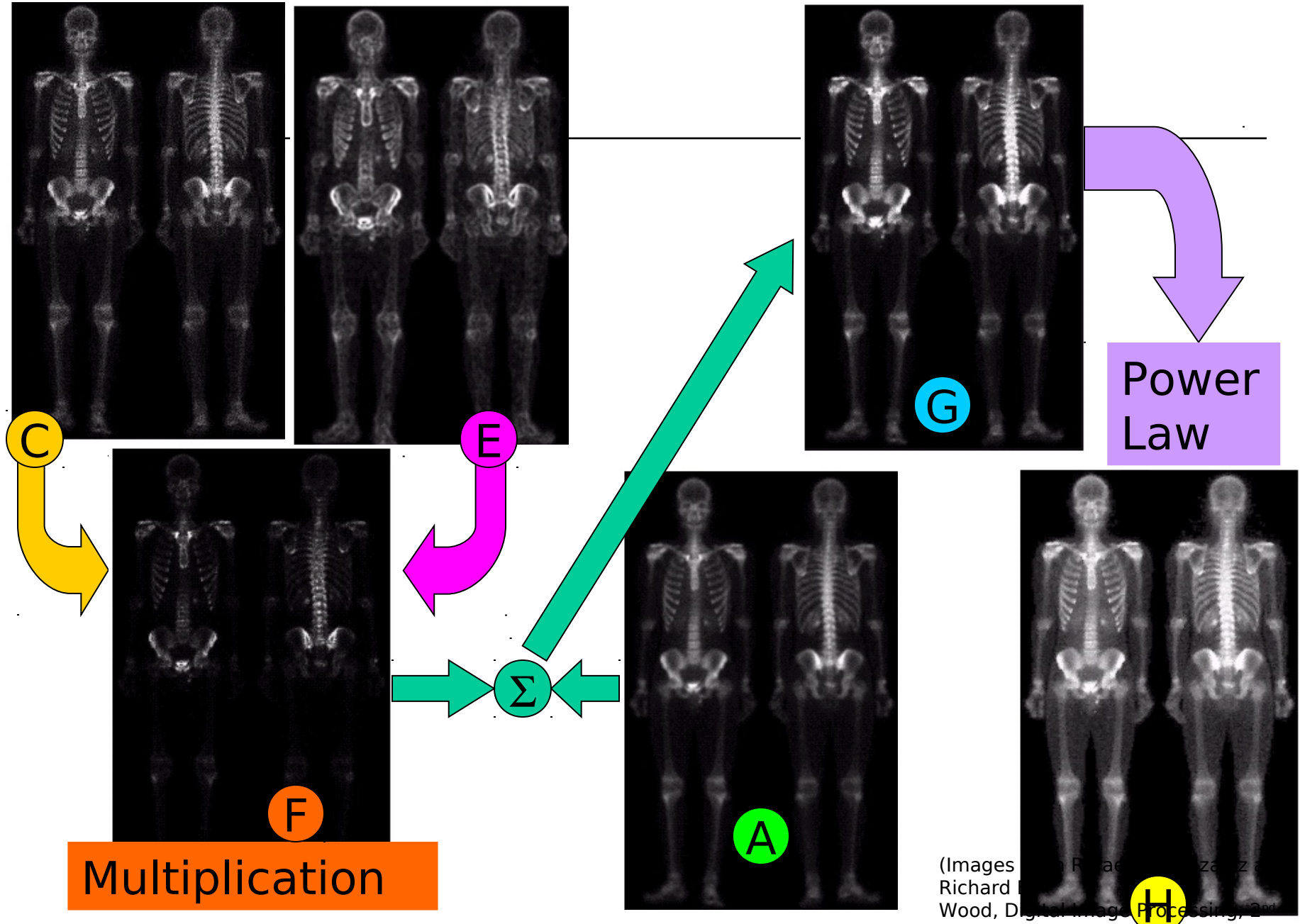


Image Enhancement in the Spatial Domain : *Mix things up !*



Combining Spatial Enhancement Methods

- An example

- Laplacian to highlight fine detail
- Gradient to enhance prominent edges
- Smoothed version of the gradient image used to mask the Laplacian image
- Increase the dynamic range of the gray levels by using a gray-level transformation